



UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

FIRST SEMESTER EXAMINATION, APRIL 2025

COURSE NO: MC/EL/GM/GL/ES/CE 361

COURSE NAME: PROBABILITY AND STATISTICS

CLASS: MC/EL/GM/GL/ES/CE III

TIME: 2 HOURS 30 MINUTES

Name: _____

Index Number: _____

INSTRUCTIONS:

The examination consists of three sections: Section A, Section B, and Section C.

Section A: Circle only the correct answers. A **PENALTY** may be applied for each incorrect answer circled.

Section B: Write down only the correct answers on the question paper. If the provided space is insufficient, you may use the answer booklet.

Section C: Answer one question in the answer booklet. I expect to read explanations and computations. **Be clear on what you are doing and why!** Ensure all responses follow the instructions for each section.

SECTION A

1. Which of the following mentioned probability distributions applies to discrete random variables?
 - a. Normal Distribution
 - b. Geometric Distribution
 - c. Exponential Distribution
 - d. Weibull Distribution

2. You are part of a team assessing the impact of industrial wastewater on a nearby river. The river stretches over 50 kilometres and passes through residential, agricultural, and industrial zones. Which sampling technique is most appropriate for accurately assessing water quality along the entire stretch of the river?
 - a. Simple Random Sampling
 - b. Convenience Sampling.
 - c. Stratified Sampling
 - d. Snowball Sampling

3. You are a materials engineer at a company that produces steel rods used in construction. Each batch produces 10,000 rods, but testing each rod for tensile strength is too costly and time-consuming. What sampling technique will be used?
 - a. Systematic sampling
 - c. Sampling random sampling

- b. Random sampling
- d. Stratified sampling
4. How many license plates can be made using 3 letters followed by 3 digits, if repetition is allowed?
 - a. 17,576,000
 - b. 26,000,000
 - c. 1,757,600
 - d. 175,760,000
 5. In a specific theodolite manufacturing process, it is known that, on average, 1 in every 100 is defective. What is the probability that the fifth item inspected is the first defective theodolite found?
 - a. 0.0010
 - b. 0.0096
 - c. 0.0091
 - d. 5518
 6. In a normal distribution, what happens to the curve if the standard deviation increases while the mean stays the same?
 - a. It becomes taller and narrower
 - b. It shifts to the right
 - c. It becomes shorter and wider
 - d. It remains unchanged
 7. Find the number of ways in which a committee of 4 can be chosen from 6 boys and 5 girls if it must consist of 2 boys and 2 girls.
 - a. 180
 - b. 160
 - c. 150
 - d. 190
 8. A remote village has 1000 vehicles with sequential registration numbers starting from 1000. Out of the total cars, 30% are without pollution clearance certificates. Further, even- and odd-numbered vehicles are operated on even- and odd-numbered dates, respectively. If 100 vehicles are chosen at random on an even-numbered date, the number of vehicles expected without pollution clearance certificate is
 - a. 15
 - b. 10
 - c. 30
 - d. 50
 9. The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000mm and 200 mm, respectively. The probability that the annual precipitation will be more than 1200 mm is
 - a. <50%
 - b. 75%
 - c. 50 %
 - d. 100%
 10. During an experiment with a particular brand of tractor engines, it was found that 80% of the engines were damaged in the first experiment. The tractor engines that survived developed a significant amount of wear and tear, so the percentage of engines that survived in any later

experiment is half that in the preceding experiment. Find the probability that an engine will survive four experiments.

- a. 0.08 b. 0.05 c. 0.45 d. 0.11

11. The hardness of a certain alloy (measured on the Rockwell scale) is a random variable X . Assume that $X \sim U(50, 70)$. Find $E(X)$

- a. 75.2 b. 65.2 c. 62.5 d. 72.5

12. For the function $f(x) = a + bx$, $0 \leq x \leq 1$, to be a valid probability density function, which one of the following statements is correct?

- a. $a = 1, b = 4$ b. $a = 0.5, b = 1$ c. $a = 0, b = 1$ d. $a = 1, b = -1$

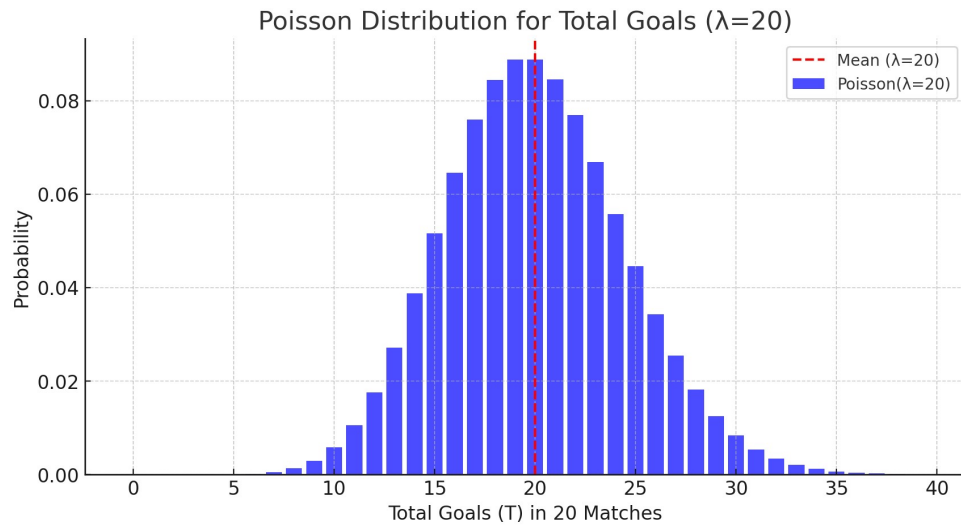
13. A drawer contains 3 white socks and 3 black socks. If I pick 2 socks at random (without replacement), what is the chance that I get a non-matching pair?

- a. 0.3 b. 0.6 c. 0.7 d. 0.9

14. Let $X \sim \text{Binomial}(5, 0.6)$ and $Y \sim \text{Poisson}(2)$ be independent. Then $P(xy = 0)$ equals:

- a. $e^{-2} \cdot (0.4)^5$
b. $e^{-2} + (0.4)^5$
c. $e^{-2} + (0.4)^5 - e^{-2} \cdot (0.4)^5$
d. $e^{-2} \cdot (0.6)^5 + (1 - e^{-2}) \cdot (0.4)^5$

15.



Given the figure above, what is the number of games required to have at least a 90% chance of scoring 30 or more goals?

a. 40

b. 38

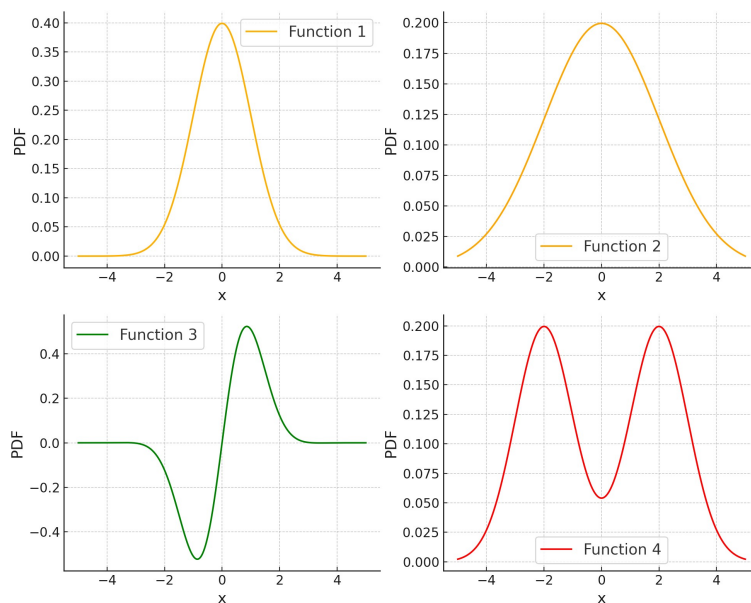
c. 37

d. 40

SECTION B

Use the following information to answer Questions 15 to 17

PROBLEM 1B:



Given the probability density functions (PDFs) in **PROBLEM 1B**:

15. **Function 1 and 2** both represents a _____ distribution with Mean of _____ and standard deviation of _____.
16. **Function 3** represents a distribution that is _____ (skewed or symmetric) and has _____ peaks.
17. **Function 4** represents a _____ distribution with _____ peaks.

Use the following information to answer Questions 18 to 20

PROBLEM 2B: If 20% of the bolts produced by a machine are bad. Determine the probability that out of 4 bolts are chosen at random.

18. What is the probability one is defective? _____
19. What is the probability none is defective? _____
20. What probability distribution does **PROBLEM 2B** follow? _____

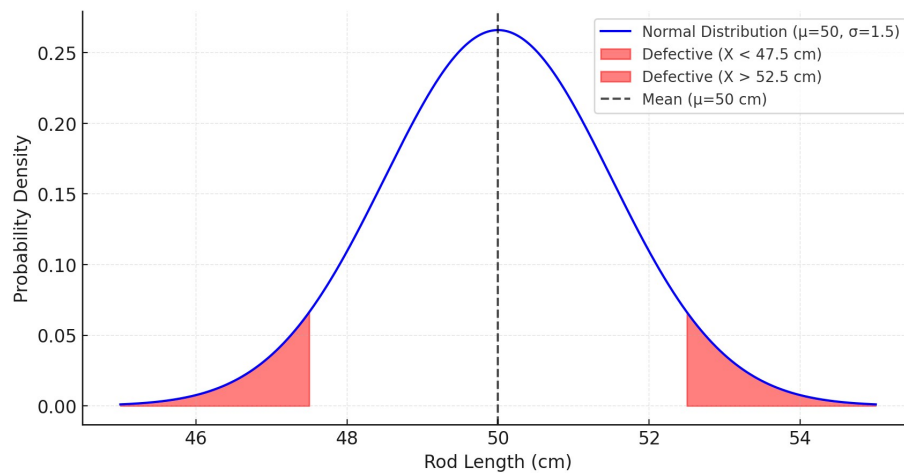
Use the following information to answer Questions 21 to 22

PROBLEM 3B: A factory has a machine that dispenses 80 mL of fluid in a bottle. An employee believes the average amount of fluid is not 80 mL. Using 40 samples, he measures the average amount dispensed by the machine to be 78 mL with a standard deviation of 2.5.

21. What is the null and research hypothesis?
- _____
- _____
22. Is there enough evidence, at a 95% confidence level, to support the idea that the machine is not working properly based on the z-score?
-

Use the following information to answer Questions 23 to 24

PROBLEM 4B: The figure above displays the quality control plot for steel rod lengths:



23. What is the probability that a steel rod has a length of less than 47 cm?

Hint: Standard Normal distribution

-
24. Construct a 99% two-sided confidence interval on the mean steel rod length given the sample 43.5, 56, 58, 34, 24, 67, 54, 30.
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Use the following information to answer Questions 27 to 29

PROBLEM 5B: Experience shows that 10% of people who make reservations for a plane trip do not show up. An airline takes 100 reservations.

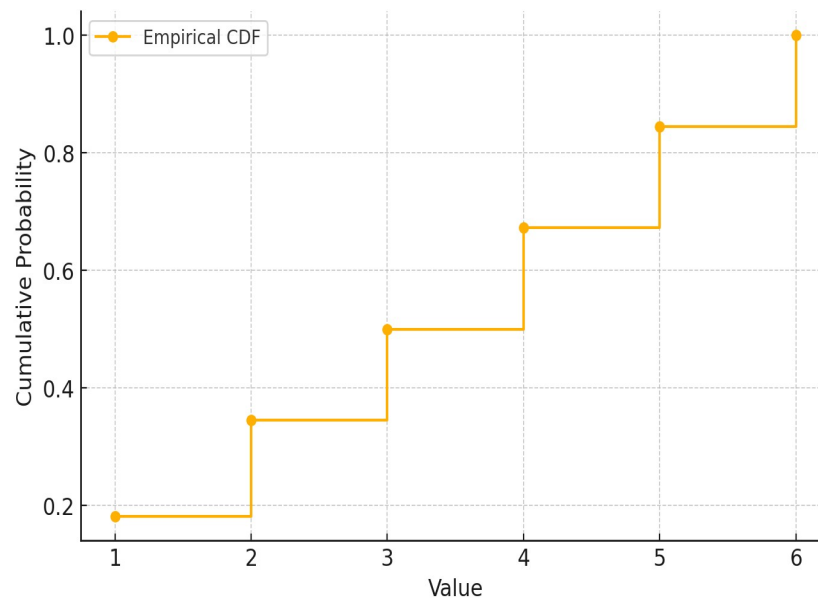
27. What is the distribution function for the number of people who will show up?
-

28. What is its mean and variance?
-

29. The plane has 90 seats. What is the probability that it will be overbooked and that no one will be able to travel?
-

Use the following information to answer Question 30

PROBLEM 6B: Based on the given Cumulative Distribution Function (CDF) graph,



30. What type of random variable does the given CDF represent, and how is it interpreted?

Hint: You may use the properties of a CDF.

SECTION C

PROBLEM 1C:

- i. The joint density for the random variable (x, y) is given by:

$$f(x, y) = 10xy^2 \quad 0 < x < y < 1.$$

Find the marginal densities $g(x)$ and $h(y)$. Derive $E(x, y)$ and $f(y|x)$.

- ii. Show that the Uniform distribution has $mean = \frac{b+a}{2}$

PROBLEM 2C:

- i. A solid propellant powers the aircrew escape system. The burning rate of this propellant is a crucial product characteristic. Specifications require that the mean burning rate must be 50 centimetres per second. We know that the standard deviation of the burning rate is 2 centimetres per second. The experimenter specifies a significance level of $\alpha = 0.05$, selects a random sample of 25 and obtains a sample average burning rate of 51.3 centimetres per second. What conclusions should be drawn?
- ii. Show that the Bernoulli distribution has $Mean (\mu) = p$

PROBLEM 3C:

A manufacturing plant uses a sensor-based fault detection system to identify defective components in an automated assembly line. The system has the following characteristics:

The system has the following characteristics:

- True Positive Rate (Sensitivity): 90%
(if a component is truly defective, the system detects it 90% of the time).
- False Positive Rate: 5%
(5% of non-defective components are incorrectly classified as defective).
- Defect Rate: 1% of all components the assembly line produces are defective.

If a randomly selected component is flagged as **defective**, use the given probabilities and the provided chart to determine the probability that the component is **defective**.

Hint: Bayes' Theorem.

Formula Sheet

Parameter	Formula
Mean (Average)	$\mu = \frac{\sum X_i}{n}$
Variance	$\sigma^2 = \frac{\sum (X_i - \mu)^2}{n}$
Standard Deviation	$\sigma = \sqrt{(\sigma^2)} = \sqrt{\frac{\sum (X_i - \mu)^2}{n}}$

Rule	Formula
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication Rule	$P(A \cap B) = P(A) P(B A) = P(B) \cdot P(A B)$
Bayes' Theorem	$P(A B) = \frac{[P(B A) P(A)]}{P(B)}$

Distribution	Formula
Binomial Distribution	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Poisson Distribution	$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$
Negative Binomial Distribution	$P(X = k) = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$
Bernoulli distribution	$P(X = k) = \begin{cases} p^k (1 - p)^{1-k} \\ 0 \end{cases}$
Geometric distribution	$P(X = k) = (1 - p)^{k-1} \cdot p$ $P(X \leq k) = 1 - (1 - p)^k$

Distribution	Formula
Normal Distribution	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$
Uniform Distribution	$f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$
Standard Normal Distribution (Z-score)	$Z = \frac{(X - \mu)}{\sigma}$

Parameter	Formula
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For Population Mean (σ known)	$CI = \bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}}$
For Population Mean (σ unknown)	$CI = \bar{x} \pm t_{\frac{\alpha}{2}, n-1} \cdot \frac{\sigma}{\sqrt{n}}$
Test Statistic for Mean	$Z = \frac{(\bar{X} - \mu_0)}{\sigma / \sqrt{n}}$